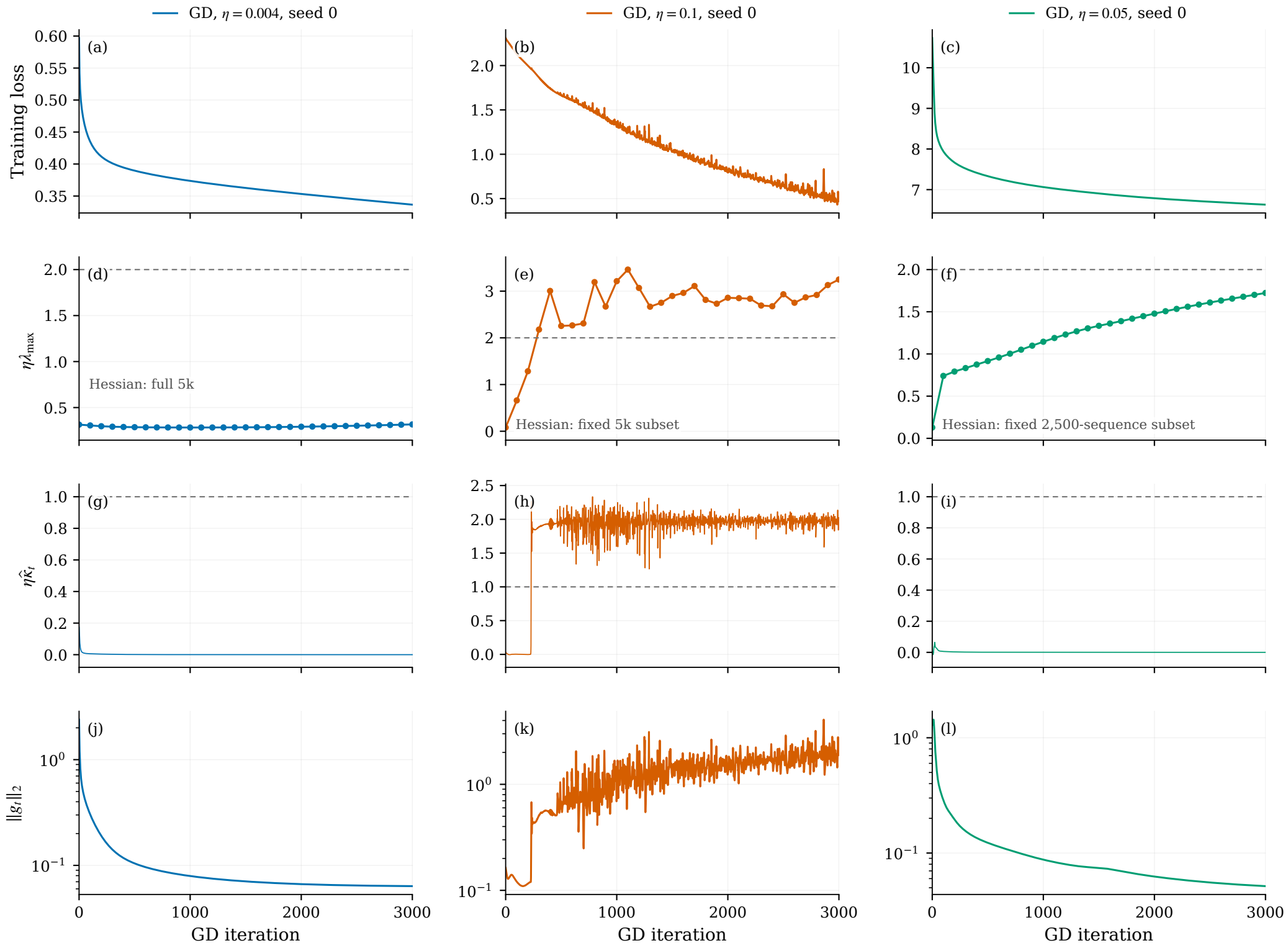

CIFAR-10 / tanh MLP
5,000 training images
CIFAR-10 / ResNet-32
50,000 training images
WikiText-2 / Transformer
2.09M training target tokens
GD, $\eta = 0.004$, seed 0
GD, $\eta = 0.1$, seed 0
GD, $\eta = 0.05$, seed 0
Training loss
(a)
(b)
(c)
$\eta\lambda_{\max}$
(d)
Hessian: full 5k
(e)
Hessian: fixed 5k subset
(f)
Hessian: fixed 2,500-sequence subset
$\eta\hat{\kappa}_t$
(g)
(h)
(i)
$\|g_t\|_2$
(j)
(k)
(l)
GD iteration
GD iteration
GD iteration